AI Fact-Checking System

Group ID
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Abstract

We demonstrate an efficient resource-optimized pipeline involving fact-checking for the climate-based claims executing in the free-tier of the Google Colab kernel runtime avoids any additional services, checkpoints of saved models, or already computed indexes. The system represents the issues of verdict predictions and retrieval of evidence below a constrained budget of compute via four staged parts: One multi-gate sparse retriever which applies a appropriately weighted reciprocal rank fusion over the views of BM25; the multi-scaling feature fusion reranker integrates similarity of sentence embedding and the shallow factual alignment cues involving numeric agreement, lexical overlapping, and the compatibility with the negation; the bounded cross-encoder selector only used on a candidate subset which is prefiltered; and TF-IDF which is a structured-organized logistic classifier which encodes the factual cue tokens and the rank of evidence as input signals that are explicit. In this respective development set, our best Colab-supporting configuration attains a solid evidence retrieval score of 0.2021, claim classification accuracy of 0.5000, harmonic mean of 0.2879, and the classifier macro-F1 score of 0.4659. The point to be noted is that we must safeguard the shallow factual cues and the evidence rank on the way via the pipeline, instead of throwing them through the concatenation of plain evidence, which uniformly enhances the performance of the classifier.

1 Introduction

Climate claims are short, compressed, and evidence-sensitive; a useful fact-checker must retrieve evidence as well as predict a label. Practical systems also face limited compute, making staged retrieval and bounded neural scoring central to the design. In this project, the system is fully self-contained at run time and generates ranked pools during notebook execution. The problem is therefore a three-way trade-off among evidence recall, verdict accuracy, and runtime.

Draft note: I collected and grouped the following papers for teammates to continue the related-work writing.

Existing work can be organized into four areas:

1. **Evidence-Based Fact Checking** FEVER frames fact checking as joint evidence retrieval and verdict prediction (Thorne et al., 2018); CLIMATE-FEVER adds climate-specific domain and retrieval challenges (Diggelmann et al., 2020).

2. **Large-Scale Evidence Retrieval** BM25 is fast and scalable (Robertson et al., 1994); reciprocal rank fusion combines ranked lists without supervision (Cormack et al., 2009). These methods provide efficient lexical recall for candidate generation.

3. **Neural Semantic Reranking** Dense embeddings help when lexical overlap is limited (Karpukhin et al., 2020; Reimers and Gurevych, 2019); cross-encoders model claim–evidence interaction directly (Nogueira and Cho, 2019). We place cross-encoder scoring after sparse prefiltering to concentrate neural computation on promising pairs.

4. **Evidence Aggregation and Factual Cue Fusion** Evidence aggregation studies passage combination before classification (Zhou et al., 2019). Hybrid systems show lexical, semantic, and shallow factual signals can be complementary (Lee et al., 2018; Padia et al., 2018).

Our design follows prior retrieval-and-reranking work and targets limited-compute execution. It first builds a broad unsupervised sparse pool from word, character, structured, and PRF gates. It then compresses this pool with MiniLM similarity plus factual features, applies a bounded cross-encoder after prefiltering, and classifies from rank-aware, cue-aware evidence text. Contributions: (i) a staged pipeline for the performance–runtime trade-off, (ii) multi-gate sparse retrieval, (iii) multi-scale reranking that fuses semantic and shallow factual signals, and (iv) diagnostics over sparse gates, fusion, evidence selection, and classifier sensitivity.

Climate misinformation has become a growing issue online, especially with claims related to climate science and global warming. Since many of these claims are presented without reliable evidence, automated fact-checking systems are becoming more important. A fact-checking system not only needs to predict whether a claim is true or false, but also needs to retrieve evidence that supports the prediction. This makes the task more difficult because both retrieval quality and classification performance affect the final result. In this project, we developed an AI fact-checking system for climate-related claims using a staged retrieval and classification pipeline. The main goal of the project was to create a system that performs well while still being efficient enough to run on free Google Colab resources. So, instead of relying on one large model, the system combines sparse retrieval, semantic reranking, factual feature fusion, and lightweight classification.

The first stage of the pipeline retrieves a broad pool of evidence using several sparse retrieval methods, including BM25 word matching, character-based retrieval, structured factual retrieval, and pseudo-relevance feedback. These retrieval sources are then combined using reciprocal rank fusion to improve evidence recall. After retrieval, the evidence candidates are reranked using MiniLM semantic embeddings together with shallow factual features such as lexical overlap, number agreement, negation matching, and the source rank. To reduce computational cost, the cross-encoder model is only applied to a smaller subset of evidence candidates instead of scoring the entire evidence corpus. Finally, the selected evidence is converted into a structured context representation and passed into a TF-IDF logistic regression classifier to predict one of four labels: "SUPPORTS", "REFUTES", "NO-ENOUGH-INFO", or "DISPUTED". Overall, the project finds out how different retrieval methods, reranking strategies, and evidence representations impact the performance of fact–checking under limited compute constraints.

2 Related Work

The dominant thought behind the process for checking the automated fact was recognized by Thorne et al. (2018), who explained the task as a verdict prediction and joint evidence retrieval across a huge document corpus data with systems needed to disproving the evidence passages or find support and classify each assertion as either "SUPPORTS", "REFUTES", or "NOT ENOUGH INFO". This method has involved a huge structure of pipeline research in which the inference of verdict, selection of candidate, and the retrieval are all treated as independent and distinct optimisable stages. The datasets that are climate-specific involve harder challenges for retrieval: Diggelmann et al. (2020) demonstrate that claims by the real-world climate includes contested framing, comparisons that are numerical, and the temporal qualifications that turn lexical matching drastically less trustworthy compared to the encyclopedic FEVER setting, inspires the multi-gate design we use and is focused on recalling.

Robertson et al. (1994) elaborates on the sparse retrieval along with BM25 being the general high-recall standard as the first stage for pipelines performing fact-checking: it scales to large corpora, needs absolutely no training, and performs way better than traditional TD-IDF at every operating point in terms of recall when measured. When many sparse systems generate ranked lists across the same corpus, the reciprocal rank fusion (Cormack et al., 2009) integrates them without needing the score calibration over the analyzers that are heterogeneous addressing a property which is important when character-level, word-level, structured-factual, and pseudo-relevance-feedback views at each produced matchless scores. These techniques are beneficial for production of candidates with high-recall but depend only on lexical matching, which inspires the summation of factual signals and semantic signals in future stages of the pipeline.

The dense passage retrieval (Karpukhin et al., 2020) embeds the documents and the queries with separate bi-encoders, allowing quick nearest-neighbour search logic over an already existing vector index and capturing evidence that the sparse gates do not catch due to the mismatch in the vocabulary. Reimers and Gurevych (2019) expand this to sentence embeddings through fine-tuning of siamese network, turning bi-encoder matching compatible to claim-evidence scoring with no training data that is task-specific; whereas the reranking stage involves an encoder for distilled sentences in this respective role. The cross-encoders (Nogueira and Cho, 2019) generate many representable scores of relevance by jointly embedding a pair of evidence-claim via complete self-attention, but their price makes the scoring of whole corpus infeasible. The general method is to ultimately use a reranker such as using cross-encoder on candidate set generated by a first stage such that the stage is cheaper, then maintaining the bounded neural step without compromising the quality of ranking, and the neural selector also proceeds in the same sequence.

The aggregation of evidence studies how several passages retrieved should be integrated before the prediction of the label. Zhou et al. (2019) represent that propagations that are graph-based, are over the nodes of retrieved evidence performing drastically well compared to treating passages as an unordered bag, showing that passage order and the relationships of inter-evidence have useful signals. The fact-checking systems which are hybrid have further noticed that semantic, lexical, and shallow factual signals are complementary instead of being mutually exclusive: Lee et al. (2018) enhances the accuracy of the claim verification by integrating lexical alignment tags with the decomposable attention models, whereas Padia et al. (2018) show that features of surface string give standalone verification signals which are comprehendable even without the use of deep neural networks. The classifier used here adheres to the line of work by embedding factual cue tokens and the evidence rank as the vocabulary of the classifier in an organized representation of TF-IDF, making the final stage execution lightweight while conserving those signals collected via the previous stages of constructed pipeline.

3 Method

3.1 Pipeline Overview

Let $\mathcal{C}$ be claims and $\mathcal{E}$ evidence. The output is a verdict and a small evidence set per claim. Since scoring all $(c, e) \in \mathcal{C} \times \mathcal{E}$ pairs is infeasible, we use a staged pipeline:

$$\mathcal{E} \rightarrow \mathcal{A}(c) \rightarrow \mathcal{B}(c) \rightarrow \mathcal{D}(c),$$

where $\mathcal{A}(c)$ is a broad sparse pool, $\mathcal{B}(c)$ a reranked context pool, and $\mathcal{D}(c)$ the submitted evidence. Cheap stages maximize recall; semantic models

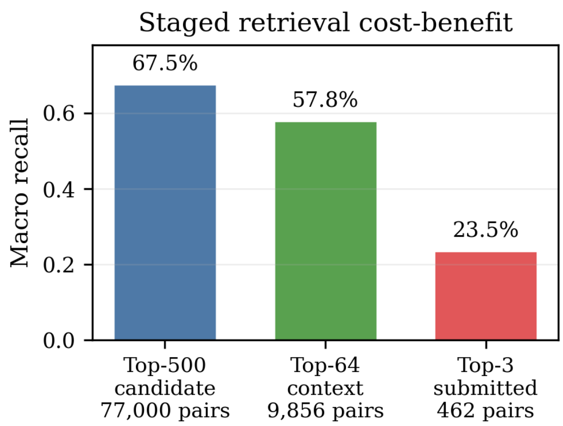

Figure 1: Staged pruning keeps recall while reducing pair count. The candidate and context pools retain most recoverable evidence by using neural scoring on progressively smaller pools; full-corpus neural scoring would require about 186 million development pairs.

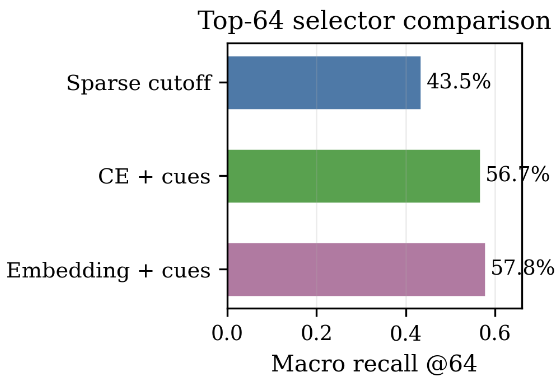

Figure 2: Embedding plus factual cues is the best top-64 selector, reaching 57.8% macro recall above sparse cutoff and cross-encoder-plus-cue variants.

operate on progressively smaller pools. The four components are multi-gate sparse retrieval, multi-scale reranking, bounded cross-encoder selection, and structured-evidence classification.

3.2 Multi-Gate Sparse Candidate Retrieval

The sparse stage maximizes recall by applying one BM25 scoring operator to multiple feature spaces. The four gates differ in analyzer $a_g(\cdot)$ and vocabulary V_g. Let $q_g(c)$ and $x_g(d)$ be the resulting count vectors for claim c and evidence document d. For each gate, the notebook builds a claim matrix Q_g and an evidence matrix X_g, reweights the evidence matrix with BM25, and computes batched sparse products $Q_g X_g^\top$ before retaining the top-ranked evidence rows per claim.

Word gate. Word terms give the direct lexical view: content-word overlap raises scores and IDF weighting emphasizes discriminative claim terms. The analyzer removes English stop words and uses both unigrams and adjacent word bigrams, so phrase matches such as scientific terms or named processes can contribute as single features.

Character gate. Character n-grams capture partial overlap, inflection, hyphenation, and spelling variation missed by exact words. This gate uses word-boundary character n-grams, making it useful for abbreviations, compounds, and morphology in climate terminology.

Structured gate. Structured cues keep numbers, years, salient capitalized terms, and negation markers. These are encoded as typed tokens such as year, number, entity-like, and negation features, so numeric or negation agreement can affect retrieval even when surrounding words differ.

Pseudo-relevance feedback gate. PRF expands the word query using high-ranked feedback evidence. If $\mathcal{F}(c)$ is the feedback set,

$$q_{\text{prf}}(c) = q_{\text{word}}(c) + \eta \sum_{d \in \mathcal{F}(c)} w(c,d) x_{\text{word}}(d),$$

where $w(c,d)$ downweights lower-ranked feedback documents. PRF adds related terms absent from the original claim while staying in the sparse word space.

Shared BM25 interaction. After feature construction, each gate applies the same BM25 interaction. For term t in document d,

$$\text{idf}_t = \log\left(\frac{N - df_t + 0.5}{df_t + 0.5} + 1\right),$$

$$\text{BM25}_{d,t} = \text{idf}_t \, \frac{(k_1+1) f_{d,t}}{f_{d,t} + k_1(1 - b + b|d|/\overline{|d|})}.$$

The inverse-document-frequency factor suppresses common features, so rare claim-specific evidence terms contribute more than corpus-wide background terms. The query side remains a count vector, giving the sparse dot product

$$s_g(c,d) = q_g(c)^\top \, \text{BM25}_g(d).$$

Thus g chooses the representation, and BM25 weights it. The notebook uses SciPy/NumPy sparse matrices, so document-frequency counts, BM25 reweighting, and query–document dot products run in optimized C/C++ kernels and vectorized array operations.

Rank fusion. Each gate produces a ranked list. The fifth sparse view is their weighted RRF fusion:

$$S_{\text{RRF}}(c,d) = \sum_{g \in G} \frac{\lambda_g}{K_{\text{RRF}} + r_g(c,d)},$$

where missing documents contribute zero. We treat λ_g as a reliability prior: stronger standalone gates receive larger weights, and complementary factual gates retain smaller weights. The structured gate contributes complementary recall in fusion, as shown by ablation; RRF also removes the need for cross-gate score calibration.

Figures 1 and 3 justify the candidate stage: fusion improves top-500 recall over any single source, and staging concentrates neural scoring on high-value candidate pools.

3.3 Multi-Scale Feature Fusion

Reranking fuses dense similarity, sparse source rank, word overlap, numeric agreement, negation compatibility, and length shape. MiniLM embeddings are mean-pooled and ℓ_2-normalized, so cosine similarity is an inner product:

$$s_{\text{emb}}(c,d) = h(c)^\top h(d).$$

The shallow factual features are summarized as

$$\phi(c,d) = [\phi_{\text{rank}}, \phi_{\text{lex}}, \phi_{\text{num}}, \phi_{\text{neg}}, \phi_{\text{len}}, s_{\text{emb}}].$$

Here ϕ_{rank} is the reciprocal sparse-source rank, ϕ_{lex} contains claim-normalized and evidence-normalized word overlap, ϕ_{num} counts shared numeric strings, ϕ_{neg} marks negation compatibility, and ϕ_{len} records normalized character-length difference. A lightweight gradient-boosted matcher is trained on gold claim–evidence pairs and sampled non-gold candidates from the training claims. It estimates $p_{\text{feat}}(c,d)$, the probability that a candidate is useful evidence under these shallow signals. The context score is

$$S_{\text{ctx}}(c,d) = \alpha \, \widetilde{s}_{\text{emb}}(c,d) + (1-\alpha) \, p_{\text{feat}}(c,d),$$

where $\widetilde{s}_{\text{emb}}$ is claim-wise normalized. Embeddings handle semantic relatedness; factual features guard against number, negation, and lexical-grounding errors.

3.4 Bounded Neural Evidence Selection

The submitted evidence stage uses a stricter operating point than the classifier context. Cross-encoders jointly encode claim and evidence, so we

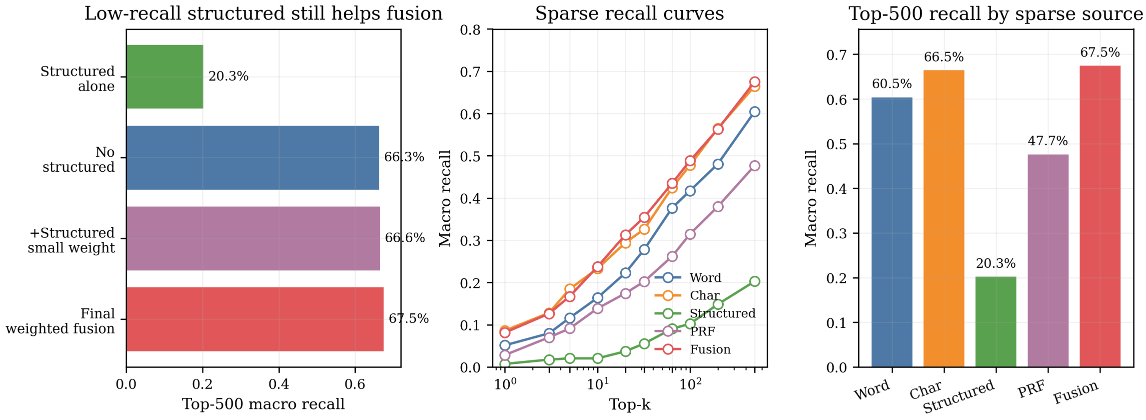

Figure 3: Sparse fusion gives the best top-500 recall and retains structured cues for complementarity. Character matching is the strongest single source; structured cues improve fused recall under controlled ablation, so the final system keeps a small structured weight.

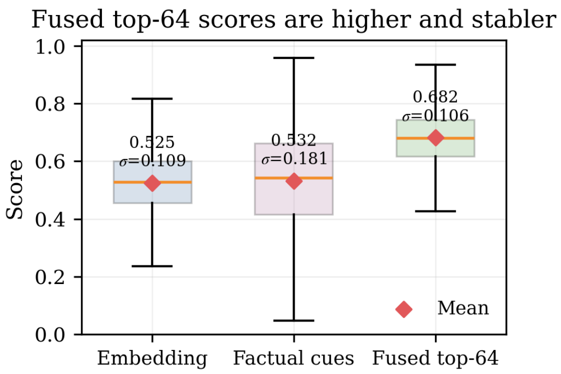

Figure 4: Top-64 fusion raises the retained-candidate score distribution and reduces dispersion. In the Colab diagnostic, fused scores have mean/std = 0.682/0.106, compared with 0.525/0.109 for embeddings and 0.532/0.181 for factual cues.

reserve them for the sparse/embedding-prefiltered pool. Candidate evidence text is truncated before tokenization, then claim–evidence pairs are scored in PyTorch batches. For each claim, raw cross-encoder logits and embedding similarities are min–max normalized over the candidate pool. The final evidence score fuses cross-encoder confidence, embedding confidence, embedding rank, and sparse source rank:

$$S_{\text{sub}}(c,d) = \beta_1 \widetilde{s}_{\text{CE}} + \beta_2 \widetilde{s}_{\text{emb}} + \frac{\beta_3}{\tau + r_{\text{emb}}} + \frac{\beta_4}{\tau + r_{\text{src}}}.$$

Coefficients are fixed. Candidates outside the cross-encoder subset remain ranked by embedding and source-rank terms, which keeps neural cost bounded and retains useful sparse candidates.

Figures 2, 4, and 5 justify multi-scale fusion: factual cues act as complementary signals that improve semantic ranking and stabilize the retained candidate pool.

3.5 Structured Evidence Classification

The classifier predicts from a bounded ordered evidence context:

$$x_c = [\text{claim}; z_1, z_1, \ldots, z_m, z_m, z_{m+1}, \ldots, z_L],$$

where the highest-ranked evidence lines are repeated to raise their TF-IDF weight. The leading evidence receives a larger text budget, while later rows are truncated so the context keeps more distinct evidence items. Each evidence line also gets a rank prefix plus cue tokens: a word-overlap bucket, number match or mismatch status, claim-number-only status, and negation compatibility. The logistic model learns weights for these cue tokens from training data. Let v_c be the TF-IDF vector; a one-vs-rest logistic model learns one binary classifier per label:

$$P(y = k \mid c) = \sigma(\theta_k^\top v_c + b_k).$$

The largest decision score gives the label. This keeps classification cheap while preserving retrieval rank and factual signals.

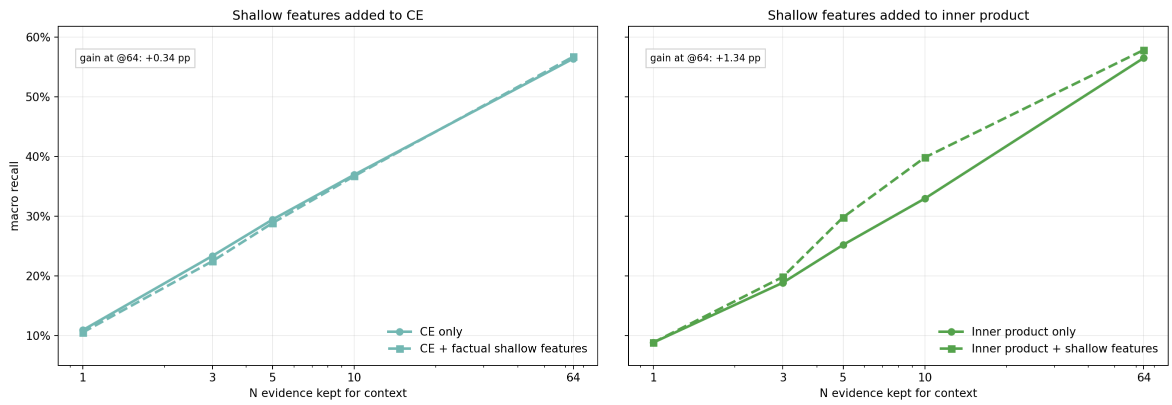

Figure 5: Factual cues preserve or improve semantic ranking. Cue gains are larger for embedding inner product and smaller for the already strong cross-encoder, supporting feature fusion over a purely semantic selector.

4 Results

The final system achieved an evidence retrieval F-score of 0.2021, claim classification accuracy of 0.5000, harmonic mean of 0.2879 and classifier macro-F2 score of 0.4679. Overall, these results showed that the system was able to balance evidence retrieval and classification performance while still remaining lightweight enough to run within Colab limitations.

The sparse retrieval experiments showed that combining multiple sparse retrieval methods improved evidence recall compared to using only a single retrieval source. Figure 3 shows that the fusion retriever achieved the highest macro recall of 67.5

The staged retrieval pipeline also helped reduce computational cost while still preserving useful evidence candidates. The sparse top-500 stage achieved 67.5

The reranking experiments showed that combining semantic embeddings with shallow factual cues improved evidence selection quality. Figure 2 shows that the top-64 selector achieved the highest macro recall of 57.8

The classifier diagnostics in Figure 7 show that the model performed best on the SUPPORTS class, while most classification errors occurred between SUPPORTS, REFUTES and NOT-ENOUGH-INFO. The DISPUTED class was the hardest to classify due to the smaller number of development examples and more ambiguous evidence relationships.

The classifier ablation experiments in Table 3 also presented that structured evidence formatting improved classification performance compared to a plain evidence sequence. Plain context achieved

Stage	F-score	Macro recall	Hit-any
Sparse top-500	0.0083	0.6754	0.9026
Context top-64	0.0517	0.5784	0.8636
Submitted top-3	0.2021	0.2355	0.4675

Table 1: Retrieval metrics from the self-generated notebook artifacts.

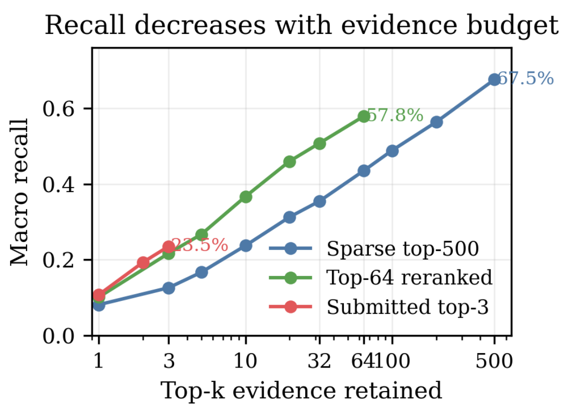

Figure 6: Recall-vs-K curves show the pruning trade-off behind the selected operating points. The sparse pool keeps broad top-500 recall, the reranked pool preserves most recoverable evidence at top-64, and the submitted top-3 is the strict evidence set used for scoring.

0.4351

Figures 6 and 7 show the final retrieval–classification behavior. Remaining classification errors mainly confuse SUPPORTS, REFUTES, and NEI. Label support is uneven, with 18 DISPUTED development examples.

Table 3 justifies the classifier input: enhanced context improves over plain context at $C = 0.25$ (0.4935/0.4544 vs. 0.4351/0.3762), and the development sweep selects $C = 0.125$ (0.5195/0.4841). The reported Colab run is Table 2: $F = 0.2021$,

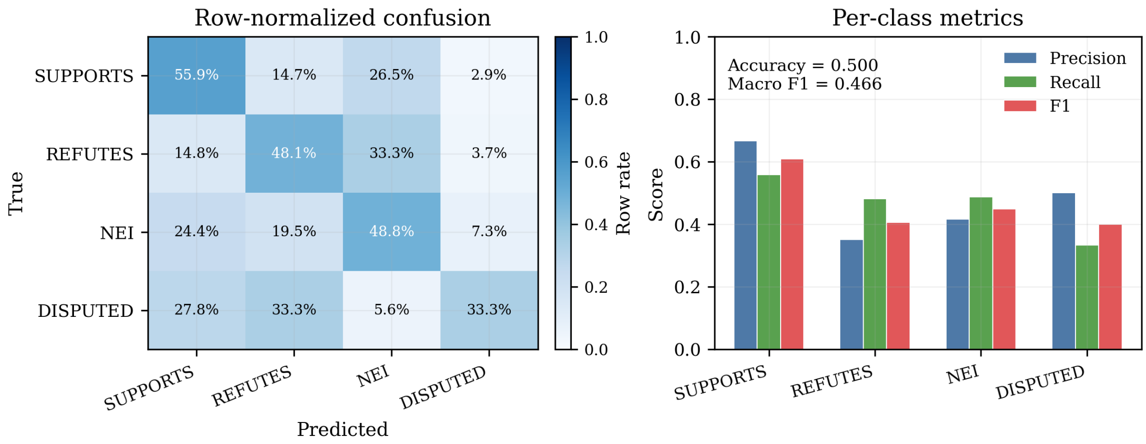

Figure 7: Classifier diagnostics. Row-normalized confusion controls for class support, and per-class precision/recall/F1 summarizes the classification report.

Metric	Value
Evidence Retrieval F-score (F)	0.2021
Claim Classification Accuracy (A)	0.5000
Harmonic Mean of F and A (H)	0.2879
Top-3 Macro Recall	0.2355
Classifier Macro-F1	0.4659

Table 2: Final Colab-reproduced development result. F, A, and $H = 2FA/(F + A)$ are official course metrics; the last two rows summarize retrieval and classification behavior.

$A = 0.5000$, $H = 0.2879$.

Context format	C	Acc.	Macro-F1
Plain	0.25	0.4351	0.3762
Rank-aware	0.25	0.4351	0.3749
Top-weighted	0.25	0.4416	0.3938
Rank-weighted	0.25	0.4351	0.3882
Enhanced	0.25	0.4935	0.4544
Enhanced	0.125	0.5195	0.4841

C	Acc.	Macro-F1
0.125	0.5195	0.4841
0.25	0.4935	0.4544
0.5	0.4740	0.4263
1.0	0.4675	0.4184
2.0	0.4416	0.3825
4.0	0.4545	0.3913
8.0	0.4481	0.3925

Table 3: Classifier ablations. Rank-aware, top-weighted, factual-cue-enhanced context improves over plain concatenation; the enhanced classifier peaks at $C = 0.125$ in the development sweep.

5 Discussion and Critical Analysis

The results showed that evidence retrieval quality had a major impact on the overall performance. Better retrieval generally led to stronger classification performance because the classifier relied heavily on receiving relevant evidence. The staged retrieval architecture worked well because it balanced retrieval recall with computational efficiency by gradually reducing the evidence pool before applying more expensive neural reranking models. This allowed the system to maintain strong retrieval coverage without requiring full corpus neural scoring.

One important finding was that sparse retrieval methods still performed strongly even when semantic reranking models were used later in the pipeline. Character-based retrieval achieved surprisingly high recall, most likely because climate-related claims often contain scientific terminology, abbreviations, and numeric information that benefit from partial lexical matching. Reciprocal rank fusion also improved retrieval performance because different sparse retrieval methods captured complementary evidence relationships.

The reranking experiments showed that semantic similarity alone was insufficient for selecting string evidence. Embedding similarity could retrieve semantically related passages, but could still rank evidence containing incorrect numbers or opposite information. Adding shallow factual cues such as lexical overlap, number agreement, and negation matching improved evidence ranking quality by helping preserve factual consistency between the claim and evidence.

The classifier ablation experiments also showed that structured evidence representation improved downstream performance. The enhanced context representation preserved evidence rank information and factual cue tokens, which helped the TF-IDF logistic regression classifier learn stronger evidence patterns compared to plain evidence.

However, the system still had several limitations. The evidence retrieval F-score remained relatively low, which showed that retrieving the exact annotated gold evidence passages was still difficult. The DISPUTED class also remained challenging due to class imbalance and more ambiguous evidence relationships with larger transformer models and more advanced end-to-end architectures.

Overall, the project showed that combining sparse retrieval, semantic reranking, factual feature fusion, and structured evidence classification could create an effective and computationally efficient fact-checking pipeline for climate-related chains.

6 Conclusion

This respective paper examined how to proportion the accuracy of claim-label, evidence recall, and the execution time for the climate claim fact-checking under the restriction of free version of Google Colab system, eliminating the concerns with respect to already computed embeddings, huge offline indexes, and costly all-pair neural scoring. The pipeline that we have developed is a strong response to answering the query via a planned framework involving a multi-gate feature fusion reranker integrating the similarity of sentence-embedding with cues of shallow factual alignment, taking into consideration a bounded cross-encoder pass for the selection of evidence, as well as an organized-evidence having a TF-IDF logistic classifier. In the respective development set, this pipeline accomplishes an evidence with F-score of 0.2021, claim accuracy classification of 0.5000, harmonic mean H-score of 0.2879, and the classifier with Macro-F1 of 0.4659, therefore surpassing the benchmarks of every component baseline beyond all metrics.

The important conclusions arise from the various trials of experiments that communicate with the research question. At first, we have the multi-gate sparse fusion achieving 67.5

The main constraint is the bottleneck between the final evidence set and the candidate gate: reaching macro recall of 67.5

7 Team Contributions

Kaitlyn- Worked on the results section, discussion and critical analysis, introduction and code. Jun- Worked on imagery, introduction, references, methods section and code Om- Worked on abstract, related work, conclusion, references and code.

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